APERIO | AUTONOMOUS EARTH INTELLIGENCE PLATFORM

AUTHORITY DISASTER INTELLIGENCE ASSESSMENT • LANDSLIDE

Location: Meppadi, Wayanad, Kerala, India • Generated: 09 Oct 2026 06:25 UTC • Status: APPROVED BY OPERATOR

Target Event	Landslide • Meppadi, Wayanad	**Risk Probability**	**85.7% (CRITICAL)**
Affected Area	16.63 km²	**Model Confidence**	95.0% Indicative
Exposed Population	37,734	**Antecedent Rainfall**	180 mm (72h: 572mm)
Terrain Slope	35° (Steep escarpment)	**SAM Scar Boundary**	Validated (7.03 km²)
Satellite Observation	Sentinel-1 SAR + Sentinel-2 MSI	**Human Verification**	APPROVED BY OPERATOR

Multi-Agent Intelligence & Evidence Progression

1. Satellite Ingestion: Copernicus Sentinel-1 C-SAR IW mode acquired observation passes over Meppadi, Wayanad. Radar backscatter penetration enabled continuous surface change mapping.
2. DEM Terrain Modeling: Horn algorithm computed local terrain slope of 35° across drainage lines.
3. SAM Spatial Segmentation: Delineated spatial impact boundary encompassing high-risk terrain.
4. XGBoost & Bayesian Sequential Update: Domain evidence likelihoods updated sequential prior belief to yield calibrated posterior risk of **85.7%**.

Response Recommendations & Relocation Planning

Designation	Capacity / Occupancy	Distance / Road Status	Hazard Exposure
Designated Relief Center (Meppadi, Wayanad)	650 / 380 (270 Available)	2.4 km • Open	**LOW RISK**
Community Relief Hall West	420 / 310 (110 Available)	4.1 km • Standby	**MODERATE**
District Camp North	800 / 220 (580 Available)	6.8 km • Clear	**LOW RISK**

Satellite Change Intelligence & Extent Quantification

Analyzed Extent	12056.04 km² (Full Granule)	**Methodology**	Visual Soil Exposure & Canopy Disturbance Delta
Baseline Affected Extent	0.12 km² (2024-06-29)	**Sensor & Product**	Sentinel-2 MSI Level-2A (Tile T43PEN)
Latest Affected Extent	781.72 km² (2024-08-13)	**Cloud Cover**	Pre: 59.04% • Post: 27.91%
Newly Affected Area	**781.6 km²**	**Net Change**	**781.6 km²** (+651333.3%)
Detection Zones	6 Delineated Zones	**Integrity Status**	**CANDIDATE_CHANGE**

Accountable Operator Review & Audit Trail

Reviewing Desk: APERIO Response Operations Desk
Operator Notes: Operator confirmed sensor convergence. Priority advisory drafted. Emergency services notified of bridge disruption.
Operational Boundary: Decision Support Output only. All evacuations require local ground confirmation.